

Access Keys and Relationships — Detailed Lecture Notes

JKSSB Computer Awareness

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29.1 Why keys and relationships matter

A database keeps data in **tables**. A table is a grid: each **record** is one row, and each **field** is one column that holds a single kind of value. When a table holds many records, values such as a name, a city, or a date can repeat, so a row cannot be recognised by those values alone. A database therefore gives each row a reliable **identifier**, called a key, and it links rows that belong together using **relationships**. Keys provide identity; relationships join tables that share data; referential integrity keeps those joins correct.

Definition (Table, record, field). A *table* stores data about one subject as rows and columns. A *record* is one complete row. A *field* is one column, holding one attribute for every record.

Term	Meaning in this lesson
Primary key (PK)	A field or set of fields that uniquely identifies each record.
Foreign key (FK)	A field that refers to the primary key of another table.
Composite key	A primary key made of two or more fields together.
Candidate key	Any field that could serve as the primary key.
Alternate key	A candidate key that was not chosen as the primary key.
Junction table	A table that links two tables in a many-to-many relationship.
Referential integrity	The rule that a foreign-key value must match an existing primary-key value.

The rest of the lesson uses these terms in one consistent sense. A **primary key** identifies a record inside its own table. A **foreign key** points from one table to another table's primary key. A **composite key** is simply a primary key built from more than one field. A **junction table** is the third table that implements a many-to-many relationship.

29.2 The primary key

A **primary key (PK)** is a field, or a set of fields, that **uniquely identifies** each record in a table. It has three firm rules. First, its value must be **unique**: no two records may share the same primary-key value. Second, it **cannot be left blank** (it cannot contain a null value), because a blank value cannot identify anything. Third, each table has **at most one** primary key, although that one key may be built from several fields.

Definition (Primary key). A field (or set of fields) whose value is unique for every record and is never blank, so it can identify each record in the table.

Example 29.2.1. In an *Employee* table, the field *EmployeeID* is a good primary key: every employee receives a different number, and no employee is stored without one. The employee's name would be a poor choice, because two employees can share a name.

Access can generate a primary key automatically with the **AutoNumber** data type, which inserts a unique number for each new record. A primary key is what makes a table the “one” side of a one-to-many relationship.

29.3 Candidate keys, alternate keys and composite keys

Any field that could serve as the primary key is called a **candidate key**. When a table has more than one candidate key, only one is selected as the primary key; the remaining candidates are called **alternate keys**. For example, a *Student* table might have both *RollNumber* and *Email* as candidate keys; if *RollNumber* is chosen, then *Email* becomes an alternate key.

A **composite key** (also called a **compound key**) is a primary key made of **two or more fields used together**. The combination is unique even when each field on its own is not. Composite keys appear most often in junction tables, where two foreign keys together form the primary key.

Definition (Composite key). A primary key that consists of two or more fields, whose combined value is unique for every record.

Example 29.3.1. An *OrderDetails* table records each product line within an order. Neither *OrderID* nor *ProductID* is unique by itself, but the pair (*OrderID* + *ProductID*) is unique for every line item, so the pair can be the primary key. This is a composite key.

A good primary key is also **stable** and **meaningless**: it should not need to change later. This is why Access often uses an AutoNumber field rather than a value such as a name or a phone number, which a person may edit.

Key	What it does
Primary key	Uniquely identifies a record; unique and not blank; at most one per table.
Candidate key	Any field eligible to become the primary key.
Alternate key	A candidate key that was not chosen as the primary key.
Composite key	A primary key made of two or more fields together.
Foreign key	A field that refers to another table's primary key; may repeat.

29.4 The foreign key

A **foreign key (FK)** is a field in one table that refers to the **primary key** of another table. The foreign key is the link that creates a relationship between the two tables. Unlike a primary key, a foreign key **may repeat** and **may be blank**; it is not unique. The table that holds the foreign key is the **child** (or related) table, and the table whose primary key is referenced is the **parent** table.

Definition (Foreign key). A field in one table whose values must match the primary-key values of another table, used to link the two tables.

Example 29.4.1. In an *Employee* table, the field *DepartmentID* is a foreign key that refers to the *DepartmentID* primary key in the *Department* table. Many employees can share the same *DepartmentID*, so the foreign key repeats; that is expected and correct.

Point	Primary key	Foreign key
Purpose	Uniquely identifies a record	Links to another table
Uniqueness	Must be unique	May repeat
Blank allowed	No	Yes
Count per table	At most one	Can be more than one

29.5 Types of relationships

A **relationship** works between two tables that share a common field — usually a primary key in one table and a foreign key in the other. Access supports three relationship types: **one-to-one**, **one-to-many**, and **many-to-many**. Relationships are created and edited in the **Relationships** window.

Type	Meaning	How it is built
One-to-one	Each record in one table matches at most one in the other	A shared field that is unique in both tables
One-to-many	One record matches many records in the other table	Primary key on the “one” side, foreign key on the “many” side (most common)
Many-to-many	Many records match many records	Two one-to-many relationships through a junction table

The rule to hold on to is simple: the “**one**” side always contributes the **primary key**, and the “**many**” side always carries the **foreign key**. A one-to-one relationship is just a special case in which both sides are unique.

29.5.1 One-to-one

In a **one-to-one** relationship, each record in the first table matches at most one record in the second table, and vice versa. It is used to split a table with many fields into smaller parts, or to keep sensitive data in a separate table that can be secured on its own. It is the **least common** of the three types.

Example 29.5.1. *An Employee table and an EmployeeConfidential table are linked one-to-one: each employee has exactly one confidential record, and each confidential record belongs to exactly one employee.*

29.5.2 One-to-many

In a **one-to-many** relationship, one record in the first table matches **many** records in the second table. The “one” side holds the **primary key**, and the “many” side holds the **foreign key**. This is the **most common** relationship type used in Access.

Example 29.5.2. *One Department has many Employees, but each Employee belongs to only one Department. The Department table is the “one” side and the Employee table is the “many” side, linked by DepartmentID.*

29.5.3 Many-to-many and the junction table

In a **many-to-many** relationship, many records in the first table match many records in the second. Access **cannot** store a many-to-many relationship directly. Instead it is split into **two one-to-many** relationships by adding a third table called a **junction table** (also known as an **associative** or **linking** table). The junction table contains a foreign key for each of the two other tables.

Definition (Junction table). *A table that connects two tables in a many-to-many relationship, holding a foreign key for each of them.*

A junction table's own primary key is usually a **composite key** formed from the two foreign keys. It may also carry extra fields that describe the link itself, such as a quantity, a grade, or a date.

Example 29.5.3. *Students and Courses have a many-to-many relationship: a student takes many courses, and a course has many students. Access implements this with an Enrollments junction table that holds StudentID and CourseID as its composite primary key, together with a Grade field.*

29.6 Creating a relationship in Access

In the Access interface, relationships are built and edited in a small number of steps. On the **Database Tools** tab, the **Relationships** button opens the **Relationships** window. The tables to be linked are added to this window, and then the **primary-key** field of one table is **dragged onto** the matching foreign-key field of the other table.

1. Open the **Relationships** window from the **Database Tools** tab.
2. Add the two tables that will be linked.
3. Drag the primary key of the “one” table onto the foreign key of the “many” table.
4. In the **Edit Relationships** dialog box, choose **Enforce Referential Integrity**.
5. Optionally tick **Cascade Update** and **Cascade Delete**, then confirm.

Example 29.6.1. *To link Departments and Employees, drag DepartmentID from the Department (parent) table onto DepartmentID in the Employee (child) table, then enforce referential integrity. Access now refuses to store an employee whose DepartmentID does not exist in the Department table.*

29.7 Referential integrity

Referential integrity is the rule that a foreign-key value must **match an existing primary-key value** in the related table (or be blank). It prevents **orphan records**, which are child rows that point to a parent row that does not exist. In Access, referential integrity is switched on in the **Edit Relationships** dialog box (opened from the **Relationships** window).

Definition (Referential integrity). *A rule that keeps the foreign key in a child table consistent with the primary key in the parent table: a child value must match an existing parent value or be null.*

When referential integrity is enforced, Access offers two related options. **Cascade Update** means that if the parent primary-key value changes, Access automatically updates the matching

foreign-key values in the child table. **Cascade Delete** means that if a parent record is deleted, Access automatically deletes the matching child records as well.

Example 29.7.1. *If a Department record is deleted and Cascade Delete is on, all Employees that belong to that Department are deleted too. Without referential integrity, Access would refuse to delete the Department while those Employees still point to it.*

29.8 Access file formats: .mdb vs .accdb

The file **extension** shows which Access format a database uses. **.mdb** (*Microsoft Database*) is the **older** Access file format, used by Access 2003 and earlier versions. **.accdb** (*Access Database*) is the **newer, default** format introduced with Access 2007 and used since. Access 2007 and later can still open an old **.mdb** file, but new databases are saved as **.accdb** by default.

Extension	Access version	Notes
.mdb	Access 2003 and earlier	The older format; supports user-level security.
.accdb	Access 2007 and later	The newer default; supports attachment, multi-valued and calculated fields.

The newer **.accdb** format added **attachment** fields, **multi-valued** fields, and **calculated** fields, and it integrates better with SharePoint and Outlook. Keep the pairing straight: **.mdb** is the *old* format and **.accdb** is the *new* format. Reversing “old” and “new” is a common exam trap.

29.9 Exam retrieval and common confusions

Six distinctions clear up most errors on this topic.

1. A primary key is **unique** and **not blank**; a foreign key **may repeat** and **may be blank**.
2. A table has **at most one** primary key, but it can hold **several** foreign keys.
3. A foreign key refers to the **primary key of another table**, not to its own table’s primary key.
4. **One-to-many** is the most common relationship; a **many-to-many** relationship is never stored directly but is built with a **junction table**.
5. A junction table commonly has a **composite primary key** made from the two foreign keys.
6. **.mdb** is the **older** Access format; **.accdb** is the **newer** format.

Example 29.9.1. *An item asks which key may contain duplicate values. Walk through the rules: a primary key must be unique and not blank, so it cannot repeat; a foreign key may repeat and may be blank; a composite key is still a primary key and so is unique as a combination. The answer is the foreign key.*

Example 29.9.2. *An item states that “a many-to-many relationship is created by adding a foreign key to one of the two tables.” This is wrong. A many-to-many relationship cannot be stored in either table directly; a **junction table** is added, and two one-to-many relationships are built through it. The wrong option has skipped the junction table.*

Quick self-check

Before the exercise set, confirm you can state each of these in one line.

- A primary key is unique, never blank, and there is at most one per table.
- A foreign key links to another table's primary key and may repeat.
- A composite key is a primary key of two or more fields.
- One-to-many is the most common relationship; the “one” side supplies the primary key and the “many” side the foreign key.
- A many-to-many relationship is always built with a junction table.
- Referential integrity keeps foreign-key values matching primary-key values and offers Cascade Update and Cascade Delete.
- .mdb is the older format; .accdb is the newer one.

Exercises

1. Define a primary key and state its two main rules about uniqueness and blank values.
2. Explain the difference between a primary key and a foreign key.
3. What is a composite key? Give one example where it is used.
4. Name the three relationship types in Access and say which is the most common.
5. Why does a many-to-many relationship need a junction table?
6. State the rule of referential integrity and name its two cascade options.
7. Distinguish the Access file formats .mdb and .accdb.

Solutions

1. A primary key is a field (or set of fields) that uniquely identifies each record. Its value must be unique — no two records may share it — and it cannot be left blank (no null value). A table has at most one primary key.
2. A primary key uniquely identifies records inside its own table and cannot repeat or be blank. A foreign key is a field that refers to another table's primary key; it may repeat and may be blank, and it creates the relationship between the two tables.
3. A composite key is a primary key made of two or more fields used together, where the combination is unique. It is used in a junction table — for example, an Enrollments table whose primary key is (StudentID + CourseID).
4. The three types are one-to-one, one-to-many, and many-to-many. The **one-to-many** relationship is the most common.
5. Access cannot store a many-to-many relationship directly. It is split into two one-to-many relationships by a junction table that holds a foreign key for each of the two tables.
6. Referential integrity means a foreign-key value must match an existing primary-key value in the related table (or be blank), which prevents orphan records. Its two options are **Cascade Update** and **Cascade Delete**.

7. `.mdb` is the older Access format used by Access 2003 and earlier. `.accdb` is the newer, default format used from Access 2007 onward, adding attachment, multi-valued and calculated fields.